

Employees

A company stores the age and salaries of its employees.

Write a program that gives:

- a) The salary of the oldest employee.
- b) The average salary of the employees.



Input

The first line of the *standard input* contains the count of employees ($1 \leq N \leq 100$). The next N lines contain the age ($1 \leq A \leq 100$) and salary ($1 \leq S \leq 2000000$) of an employee each.

Output

The first line of the *standard output* should contain the salary of the oldest employee (if there is more than one solution, give the one with the smallest index).

The second line of the *standard output* should contain the average salary of the employees (rounded to two decimals).

Example

Input

```
6
42 150000
33 100000
25 600000
66 80000
25 200000
42 800000
```

Output

```
80000
321666,67
```